

Static Routing:

Static routing is a type of network routing technique where routes are manually configured and maintained by a network administrator. This method involves specifying the path that data packets should take to reach a specific destination, without relying on dynamic routing protocols.

Key Features of Static Routing

- Manual Configuration**
- No Overhead**
- Simplicity**
- Predictable Behavior**

Advantages of Static Routing:

Control: Administrators have full control over the routing paths.

Security: Less susceptible to certain types of attacks compared to dynamic routing protocols.

Reduced Resource Usage: No CPU or memory overhead for route calculations or updates.

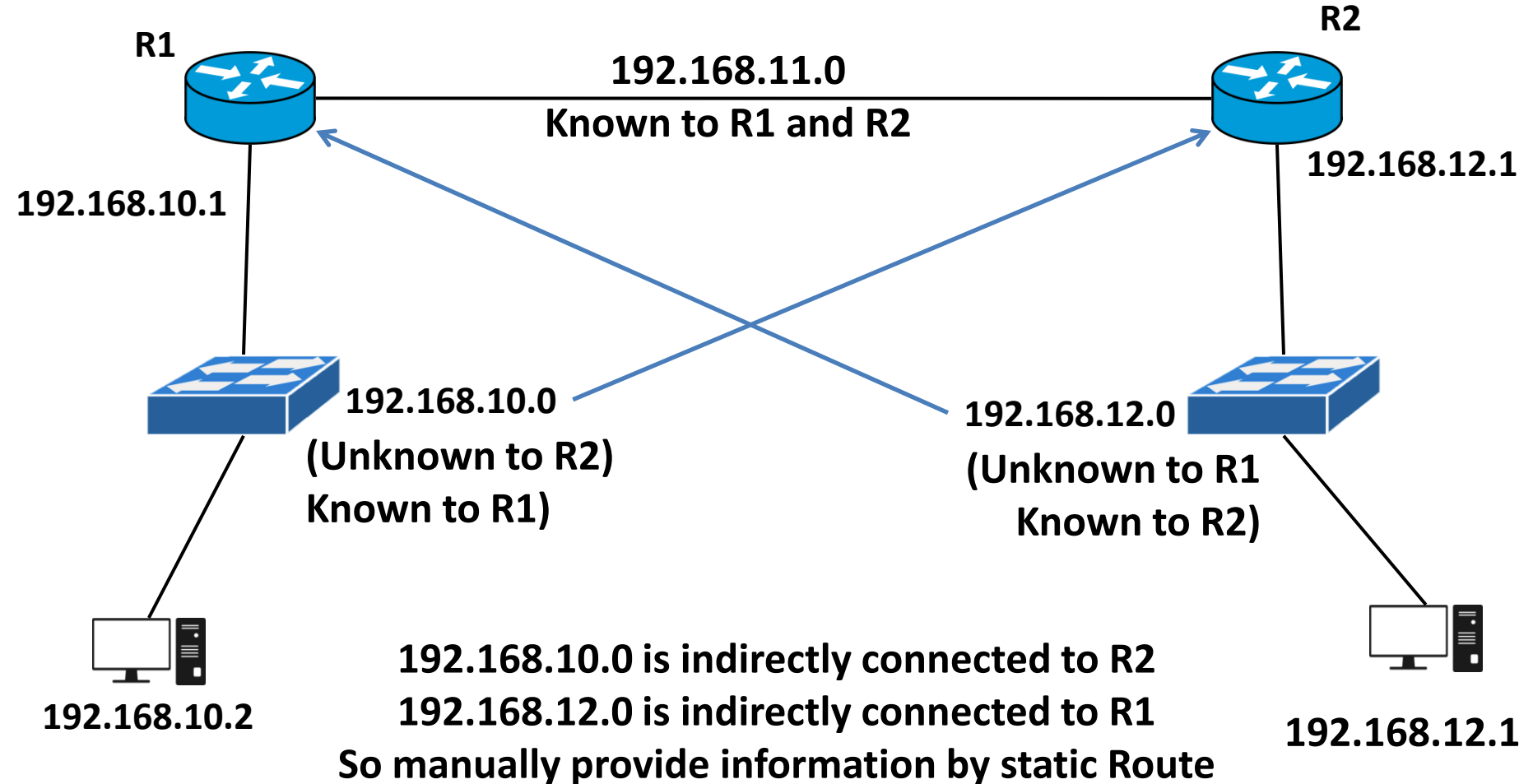
Disadvantages of Static Routing:

Scalability: Not practical for large networks due to the administrative overhead of maintaining routes.

Lack of Redundancy: Does not automatically adapt to network changes or failures, potentially leading to downtime.

Error-Prone: Manual configuration can lead to human errors, resulting in misconfigured routes.

Why Need Routing:



Static Route Configuration:

```
Router> enable
```

```
Router# configure terminal
```

Command for Config:

```
Router(config)# ip route <destination_network>  
<subnet_mask> <next_hop_ip>
```

Example:

```
Router(config)# ip route 192.168.2.0 255.255.255.0  
192.168.1.2
```

```
Router# write memory
```

```
Router# show ip route
```

Next hop IP:

